EXPERIMENT 11
GENERIC CONVECTION–DIFFUSION TRANSPORT

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1. Aim

To compare scalar-temperature transport in three prescribed flow configurations: forced flow over a flat plate, forced flow through a channel, and a natural-convection surrogate represented by a prescribed recirculating velocity field. The comparison is based on temperature-field shape, extracted temperature profiles, outlet or mean temperature, wall heat flux and iterative residual.

2. Governing equation

The notebook solves a two-dimensional transient convection–diffusion equation for a scalar temperature field T(x,y,t):

$$\partial T/\partial t + u\,\partial T/\partial x + v\,\partial T/\partial y = \alpha\,(\partial^2 T/\partial x^2 + \partial^2 T/\partial y^2)$$

Here, u and v are the prescribed velocity components in the x and y directions, and α is the scalar diffusivity. The Peclet number is Pe = 200 for the lab preset and α = 1/Pe. The convection terms are evaluated using first-order upwind spatial derivatives, while diffusion is evaluated using a central-difference Laplacian. The field is advanced through explicit transient iterations.

3. Flow cases and boundary conditions

3.1 Forced flat-plate case

The computational domain is $0 \le x \le 5$ and $0 \le y \le 1$. A prescribed external-flow velocity is used, with $u = \tanh[y/\delta(x)]$ and $v = 0$, where $\delta(x) = 0.035 + 0.22\sqrt{(\max(x,dx)/L)}$. The lower wall is heated to $T = 1$ from $x/L \ge 0.25$; the unheated portion is $T = 0$. The inlet is $T = 0$, the upper boundary is $T = 0$, and the outlet uses a zero-gradient condition.

3.2 Forced pipe/channel case

The domain is $0 \le x \le 5$ and $0 \le y \le 1$. The prescribed parabolic channel velocity is $u = 6\eta(1-\eta)$, where $\eta = y/H$, and $v = 0$. The inlet is $T = 0$, both channel walls are maintained at $T = 1$, and the outlet uses a zero-gradient condition.

3.3 Natural vertical-plate surrogate

The domain is $0 \le x \le 1$ and $0 \le y \le 1$. A prescribed divergence-free recirculating velocity field is applied: $u = -\sin(\pi x/L)\cos(\pi y/H)/H$ and $v = \cos(\pi x/L)\sin(\pi y/H)/L$ for the unit-height, unit-length domain. The initial temperature is $T = 1 - x/L$. The left boundary is $T = 1$, the right boundary is $T = 0$, and the top and bottom boundaries use zero-normal-gradient conditions. This is a prescribed-flow surrogate, not a coupled Boussinesq natural-convection calculation.

4. Numerical method and parameter set

The solver performs explicit finite-volume-style transient iterations on a uniform Cartesian grid. At each iteration it evaluates upwind convection, central-difference diffusion, updates interior temperatures, clips the field to the interval −0.05 to 1.05, reapplies boundary conditions and evaluates the RMS change in the interior field.

Run mode evidenced by supplied log	quick (8 iterations; workflow check)
Case selection	all three cases
Grid evidenced by quick preset	nx = 25, ny = 21
Peclet number, Pe	100 (quick preset)
Diffusivity, $\alpha = 1/Pe$	0.01
Maximum iterations	8
Output interval	8 iterations; logs at iteration 1 and 8
Domain: forced flat plate	5 × 1
Domain: forced channel	5 × 1
Domain: natural plate surrogate	1 × 1
Time-step selection	$dt = \min[0.35 \min(dx/\max\|u\|, dy/\max\|v\|), 0.20 \min(dx^2,dy^2)/\alpha]$

Parameter clarification: the notebook cell shown in the prompt displays lab defaults of 121 × 61, Pe = 200, 3000 iterations and output_every = 300. However, the actual supplied log ends at iteration 8 and matches the QUICK_ARGS preset [all, 25, 21, 100, 8, 8]. This report therefore records the settings evidenced by the result log, not the unexecuted lab defaults.

5. Residual definition and convergence history

The code defines the residual as the root-mean-square change in interior temperature between consecutive iterations:

$$\text{Residual} = \sqrt{\text{mean}[(T^{n} - T^{n-1})^{2}]} \text{ over interior grid points}$$

The following values are copied from the supplied solver output. Only iteration 1 and iteration 8 were recorded in the quick run; therefore, intermediate residual values are not available.

Case	Iteration	Residual	Outlet / mean T	Average wall flux
Forced flat plate	1	4.059061×10^{-2}	2.427478×10^{-3}	16.00000
Forced flat plate	8	8.701541×10^{-3}	1.853125×10^{-2}	8.391536
Forced pipe/channel	1	6.308611×10^{-2}	5.555556×10^{-3}	16.11111
Forced pipe/channel	8	1.395576×10^{-2}	4.207700×10^{-2}	8.449362
Natural vertical plate surrogate	1	7.249832×10^{-3}	0.500000	1.000000
Natural vertical plate surrogate	8	6.989738×10^{-3}	0.500000	1.056715

Residuals decrease from iteration 1 to iteration 8 in all three cases. The natural-plate surrogate changes only slightly, whereas the forced cases show larger residuals at iteration 8. None of these quick-run residuals demonstrates convergence to a steady solution; the run is too short to claim convergence.

6. Quantitative case comparison

The iteration-8 values are the final values available in the supplied quick-run output:

Case	Outlet or mean temperature	Average wall flux	Final residual
Forced flat plate	0.01853125	8.391536	8.701541×10^{-3}
Forced pipe/channel	0.04207700	8.449362	1.395576×10^{-2}
Natural vertical plate surrogate	0.500000	1.056715	6.989738×10^{-3}

The natural-plate surrogate has the largest reported mean temperature (0.5), because its reported metric is the domain-interior mean and its initial/boundary temperature field spans the range from 1 at the left to 0 at the right. The forced pipe/channel case has a higher outlet temperature (0.04208) than the forced flat-plate case (0.01853). The forced cases have much larger reported average wall fluxes (about 8.4) than the natural surrogate (about 1.06). These metrics are defined differently by case in the code: the forced cases use a velocity-weighted outlet temperature, while the natural case uses an interior mean temperature. They should not be interpreted as perfectly equivalent physical measurements.

7. Extracted temperature profiles

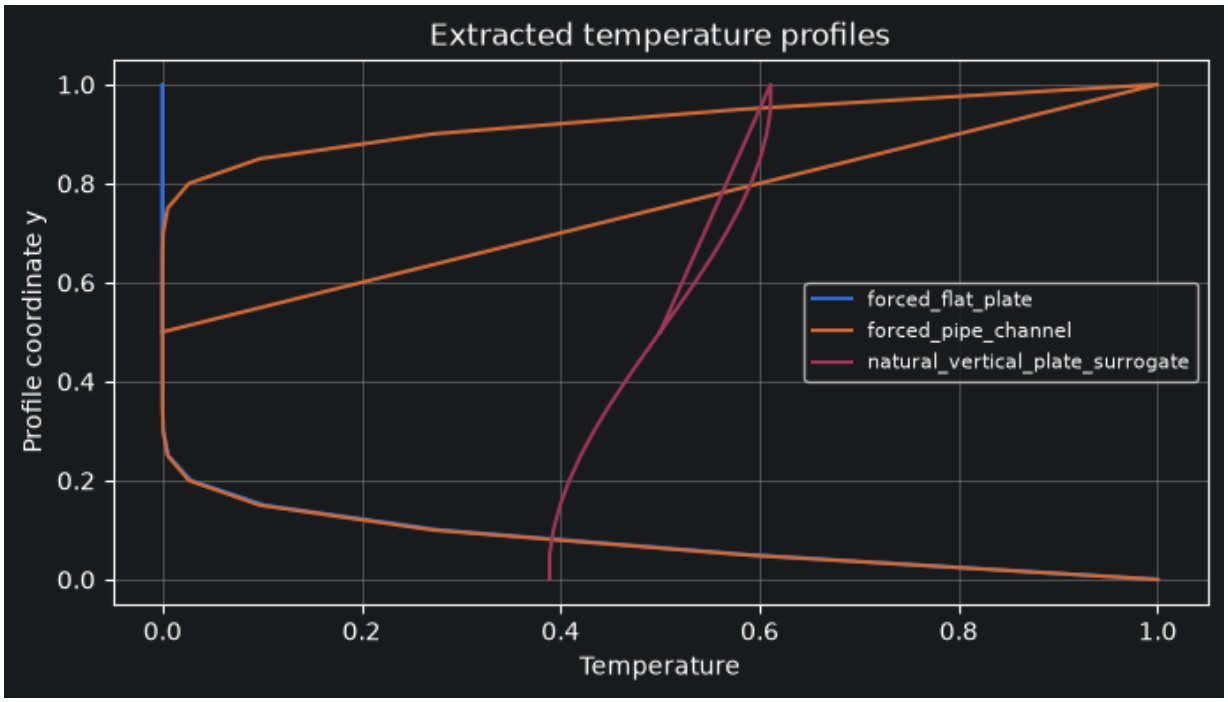


Figure 1. Extracted temperature profiles for the three prescribed-flow cases.

The forced flat-plate profile shows a steep temperature change close to the heated lower wall, followed by a region at lower temperature farther from the wall. This indicates a relatively thin thermal boundary layer in the displayed quick-run result. The forced channel profile is influenced by heating from both walls and the parabolic channel velocity; the profile is not symmetric in the plotted quick-run output because the temperature field has undergone only eight iterations from its initial condition. The natural-plate surrogate profile varies smoothly across the domain, reflecting the imposed left-to-right temperature difference and recirculating prescribed velocity field.

8. Case comparison plot

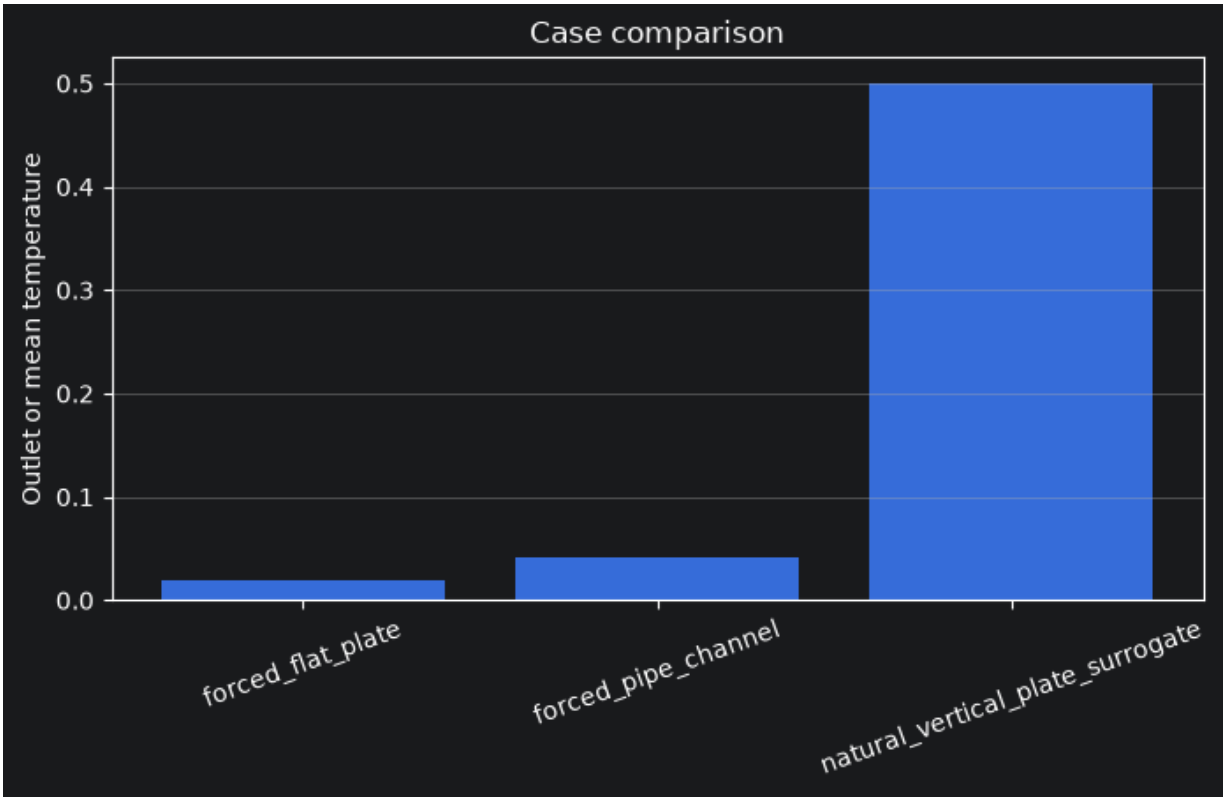


Figure 2. Comparison of outlet or mean temperature at the final recorded iteration.

The natural vertical-plate surrogate reaches a mean temperature of 0.5, substantially greater than the forced flat-plate and forced channel outlet values. The forced channel outlet value is approximately 0.0421, compared with approximately 0.0185 for the forced flat plate.

9. Inline temperature-field visualization

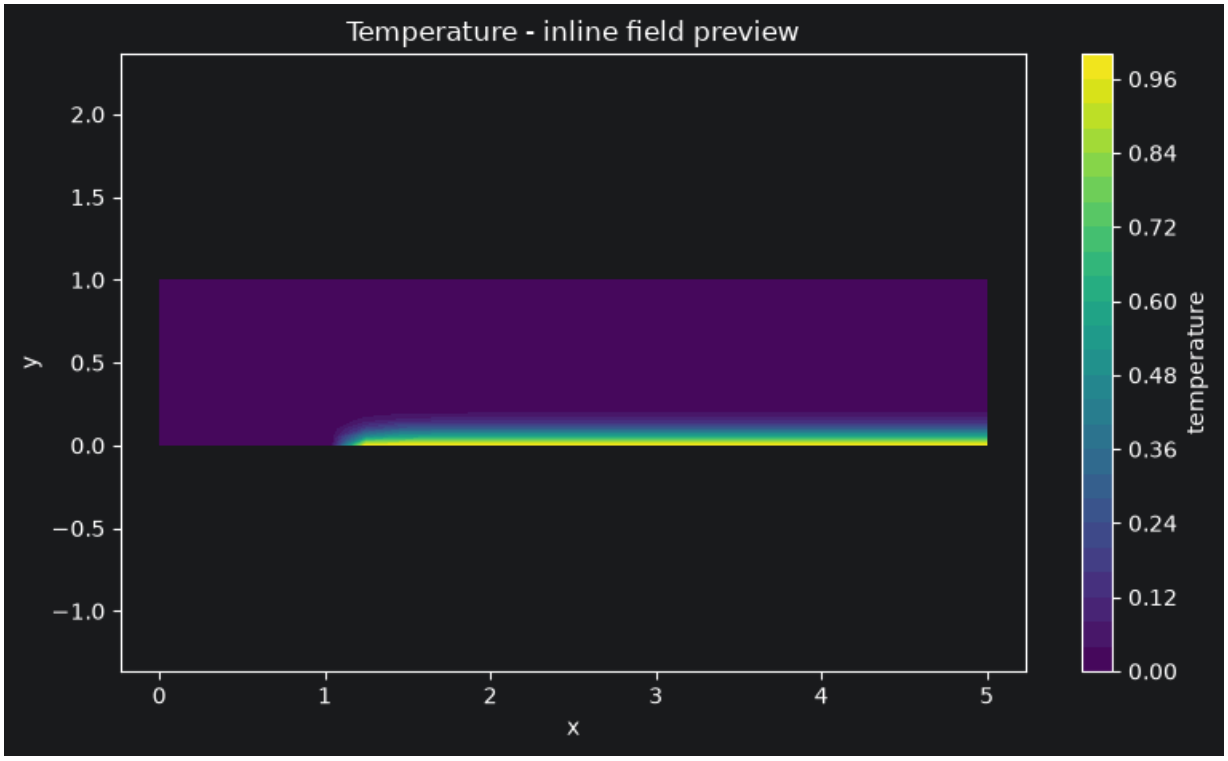


Figure 3. Inline temperature contour for the forced flat-plate case. The x-axis is the streamwise coordinate, the y-axis is the wall-normal coordinate, and the colour bar represents temperature.

The contour shows the heated region along the lower boundary, beginning downstream of the unheated leading section. Temperature is highest near the heated wall and decreases away from it. The thermal layer grows downstream, which is consistent with convection and diffusion acting together over a heated flat plate.

10. Observed numerical behaviour versus physical interpretation

Numerical observations from the supplied results

- At iteration 8, the forced flat-plate outlet temperature is 0.01853125 and its average wall flux is 8.391536.
- At iteration 8, the forced channel outlet temperature is 0.04207700 and its average wall flux is 8.449362.
- At iteration 8, the natural-plate surrogate mean temperature is 0.500000 and its average wall flux is 1.056715.
- The residual decreases between iterations 1 and 8 in all three cases.
- The temperature contour for the flat plate shows a heated near-wall layer that extends downstream.

Physical interpretation

Convection transports temperature in the direction of the local velocity field, while diffusion spreads temperature gradients across the flow. In the forced flat-plate case, the heated wall produces a thermal boundary layer that thickens downstream. In the channel case, both walls are heated and the parabolic velocity field carries the warmed scalar toward the outlet. In the natural-convection surrogate, the prescribed recirculating velocity redistributes the scalar, but buoyancy is not calculated from temperature; therefore, the result represents transport in an imposed circulation rather than a fully coupled natural-convection process.

11. Limitations of the teaching model

- The quick run uses a coarse 25 × 21 grid and only eight iterations; its values are preliminary and not suitable as final converged lab results.
- The velocity fields are prescribed and are not updated in response to the temperature field.
- The natural-convection case is not a coupled Boussinesq solution; buoyancy and momentum equations are not solved together with the energy equation.
- The method uses explicit time stepping and first-order upwind convection, which can introduce numerical diffusion.
- The temperature is clipped to −0.05 to 1.05, which can limit extreme numerical values.
- The outlet temperature metrics differ: velocity-weighted outlet temperature is used for forced cases, while an interior mean is used for the natural surrogate.
- Only iteration 1 and iteration 8 are available in the supplied log, so a complete residual curve cannot be reconstructed.
- The supplied results do not provide a separate quantitative boundary-layer-thickness calculation; thickness is discussed qualitatively from the profile and contour plots.

12. Conclusion

The experiment compares temperature transport in forced flat-plate, forced channel and natural-convection-surrogate flow fields. In the supplied eight-iteration quick run, the forced channel case gives a higher outlet temperature than the forced flat plate, while the natural-plate surrogate gives a mean temperature of 0.5. The forced cases show larger average wall-flux values than the natural surrogate. The flat-plate contour illustrates the development of a heated near-wall thermal layer. Residuals decrease over the recorded iterations, but the run is not sufficiently long to establish convergence. The results should therefore be treated as a workflow demonstration; the prescribed lab run is required for final quantitative submission.